

“Android is 18 Months Away From Embedded Space”



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Bharat Shah, vice president and country manager, Janardhan Prabhakar, national sales manager, and Nigel Forrester, market development manager, Emerson Network Power (Embedded Computing) spoke with EFY's Dilin Anand about the trends in industrial automation and company's R&D in India

What's up in the industrial automation market?

Automation itself is a very broad field. If you are making a robot controller, for example, you will require different interfaces that are specific to the application because of the greater level of integration. But even with all this miniaturisation, when we walk into a factory, electricians still require screw terminals or connectors that they can physically see. So although miniaturisation in electronics is good, you still have to see how these products are going to be used in the end environment.

Can Android serve as a suitable platform for this market as well?

As Android is still focused towards the mobile handset space, right now it is the companies concerning in-flight entertainment and medical equipment which have started to look at Android. We believe that Android is still 18 months away from becoming stable enough for the embedded space. To this day, customers are using Linux

variants, real-time operating systems, etc.

What are the technical challenges in designing a computer-on-module?

Computer-on-modules (COMs) are small single-board computers, split into two modules. One is the main module and the other is the baseboard. COM Express, for example, currently defines basic and compact-sized modules that have a size of 125×95 mm² and 95×95 mm², respectively. Fitting a processor, memory, flash and interfaces within such a small space requires careful layout with respect to environmental conditions such as temperature.

We carry out simulations followed by extensive validation testing to ensure that our COM solutions, including a heat-sink or heat spreader, meet our stated specifications. In addition, COMs are designed and rigorously tested to ensure that the clock, power, reset and memory signals have sufficient margin to ensure consistent operation over the lifetime of the product.